

Expanded Unsupervised Learning Algorithm Family Map

1. Clustering Methods

Partition-Based Clustering:

- K-Means
- K-Medoids (PAM)
- CLARA
- CLARANS
- Fuzzy C-Means

Hierarchical Clustering:

- Agglomerative (Bottom-Up)
- Divisive (Top-Down)
- Single Linkage
- Complete Linkage
- Average Linkage
- Weighted Linkage
- Ward's Method
- Centroid Linkage

Density-Based Clustering:

- DBSCAN
- HDBSCAN
- OPTICS

Model-Based Clustering:

- Gaussian Mixture Models (GMM)
- Expectation Maximization (EM)
- Bayesian Gaussian Mixtures

Graph/Spectral Clustering:

- Spectral Clustering
- Community Detection (Louvain, Leiden)

2. Association Rule Learning

- Apriori Algorithm
- FP-Growth
- ECLAT
- AIS Algorithm
- SETM Algorithm

Common Measures:

- Support
- Confidence
- Lift
- Leverage
- Conviction

3. Dimensionality Reduction

Linear Methods:

- PCA
- Kernel PCA
- Incremental PCA

- Sparse PCA
- Factor Analysis
- Independent Component Analysis (ICA)
- Singular Value Decomposition (SVD)
- Truncated SVD (LSA)

Nonlinear / Manifold Learning:

- t-SNE
- UMAP
- Isomap
- Locally Linear Embedding (LLE)
- MDS (Multidimensional Scaling)
- Laplacian Eigenmaps

4. Anomaly / Outlier Detection

Statistical Methods:

- Z-Score Method
- Standard Deviation Method
- Modified Z-Score
- IQR Method
- Grubbs' Test

Distance / Density Methods:

- Local Outlier Factor (LOF)
- k-NN Outlier Detection
- DBSCAN-based Detection

Tree-Based Methods:

- Isolation Forest
- Extended Isolation Forest

Model-Based Methods:

- One-Class SVM
- Elliptic Envelope
- Autoencoder-based Anomaly Detection

5. Performance Evaluation

Clustering Metrics:

- Within-Cluster Sum of Squares (WCSS)
- Silhouette Score
- Davies-Bouldin Index (DBI)
- Dunn Index
- Calinski-Harabasz Index
- Inertia
- Rand Index (RI)
- Adjusted Rand Index (ARI)
- Normalized Mutual Information (NMI)
- Adjusted Mutual Information (AMI)
- Homogeneity Score
- Completeness Score
- V-Measure

Dimensionality Reduction Metrics:

- Explained Variance Ratio (PCA)
- Reconstruction Error
- Trustworthiness

- Continuity

Association Rule Metrics:

- Support
- Confidence
- Lift
- Conviction
- Leverage

Anomaly Detection Metrics:

- Precision
- Recall
- F1-Score
- ROC-AUC
- PR-AUC